Linear Algebra and Its Applications

Section 1.S: Chapter 1 Supplementary Exercises

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

1.S Chapter 1 supplementary exercises 3

Exercise 1.S.1: Twenty-six statements and their logical pitfalls 3

Exercise 1.S.2: Every possible solution set of one equation 4

Exercise 1.S.3: Three configurations of planes 4

Exercise 1.S.4: Why the square size matters 5

Exercise 1.S.5: Classify two parameter-dependent systems 5

Exercise 1.S.6: Restate one consistency question three ways 5

Exercise 1.S.7: Restate consistency for every right-hand side 5

Exercise 1.S.8: Echelon patterns for spanning columns 6

Exercise 1.S.9: Decompose a vector along two different lines 6

Exercise 1.S.10: Infer existence and uniqueness from the diagram 7

Exercise 1.S.11: A nonechelon matrix with a line of zero-images 7

Exercise 1.S.12: A nonechelon matrix with a plane of zero-images 7
Exercise 1.S.13: A prescribed null vector determines the missing entries 7
Exercise 1.S.14: Parameter values for two-vector independence 7
Exercise 1.S.15: Parameters in triangular-looking vector lists 8
Exercise 1.S.16: Use the preceding-column characterization 8
Exercise 1.S.17: Adjoin a vector outside the old span 8
Exercise 1.S.18: Add one basis vector to another 8
Exercise 1.S.19: Three collinear points give dependent position vectors 9
Exercise 1.S.20: Reverse an input using homogeneity 9
Exercise 1.S.21: A coordinate-plane reflection 9
Exercise 1.S.22: Surjectivity forces injectivity for this square size 9
Exercise 1.S.23: A Givens rotation in two coordinates 9
Exercise 1.S.24: Zero one coordinate without changing the middle one 9
Exercise 1.S.25: A building design needs whole, nonnegative floor counts 10

1.S Chapter 1 supplementary exercises

These exercises combine consistency, parametrization, independence, and transformations. Keep existence and uniqueness separate: a missing variable pivot creates freedom only in a *consistent* system. When a problem describes physical quantities such as floors, solve the real linear equations first, then impose the required integer and nonnegative restrictions.

Exercise 1.S.1: Twenty-six statements and their logical pitfalls

(a) **False.** Reduced echelon form is unique; ordinary echelon form is not. The one-by-one matrices [1] and [2] are row equivalent and both in echelon form, but are different.

(b) **False.** For $n = 1$, the equation $0x = 0$ has infinitely many solutions, more than one. A square system can have zero, one, or infinitely many solutions.

(c) **True.** If $Ax = Ay = b$ and $x \neq y$, then $A(x + t(y - x)) = b$ for every real t. Different t give distinct solutions because $y - x \neq 0$.

(d) **False.** No free variables guarantees at most one solution, not consistency. The equations $x = 0$ and $x = 1$ have no free variable and no solution.

(e) **True.** Elementary row operations are reversible, so each operation preserves the solution set in both directions.

(f) **True.** If $Ax = Ay = b$ with $x \neq y$, then $A(x - y) = 0$ with $x - y \neq 0$. Thus the homogeneous system has both zero and a distinct nonzero solution.

(g) **False.** Every A has a consistent system for $b = 0$. For the zero 2×2 matrix the columns span only $\{0\}$, not $\mathbb{R}^2$.

(h) **False.** Every matrix has a reduced echelon form, including inconsistent augmented matrices. The single equation $0x = 1$ is already a counterexample.

(i) **True.** Each row-equivalence class has one reduced echelon form. Reducing either member therefore gives the same matrix.

(j) **False.** $Ax = 0$ always has the trivial solution, including when free variables exist. Take $A = [0]$.

(k) **True.** Consistency for every $b \in \mathbb{R}^m$ means a pivot in every one of the m rows. There are exactly m pivots and thus m pivot columns.

(l) **False.** Row pivots imply existence for every target, not uniqueness. The matrix [1 0] has a pivot in every row, but $x_1 = b$, with x_2 arbitrary, gives infinitely many solutions.

(m) **True.** In a square $n \times n$ matrix, n pivots occupy every column and every row. Reduced pivots are ones with zeros elsewhere in their columns, so the reduced matrix is I_n.

(n) **True.** Both matrices reduce to I_3. Reduce A to I_3, then reverse the operations reducing B to I_3 to reach B.

(o) **True.** The first system supplies a nonzero z with $Az = 0$. If $Ay = c$, then $A(y + tz) = c$ for every real t, giving infinitely many distinct solutions of the second system.

(p) **True.** Row-equivalent matrices have the same number of pivots. If A spans $\mathbb{R}^m$, its m pivots fill all rows; the same is true of B, so its columns also span $\mathbb{R}^m$. This does not assert equality of their actual column vectors.

(q) **False.** The list $e_1, e_2, e_1 + e_2$ in $\mathbb{R}^3$ has no proportional pair but has the nontrivial relation $e_1 + e_2 - (e_1 + e_2) = 0$.

(r) **True.** Three vectors in $\mathbb{R}^2$ cannot be independent: a 2×3 matrix has at most two pivots. Thus the assumed independence excludes that ambient space.

(s) **False.** Four columns supply at most four pivots, not the five row pivots required to span $\mathbb{R}^5$.

(t) **True.** $-u = (-1)u + 0v$, a permitted linear combination.

(u) **False.** Let $u = e_1$, $v = 2e_1$, $w = e_2$ in $\mathbb{R}^2$. All are nonzero, but the first two span only the horizontal axis, which does not contain w.

(v) **False.** Take $u = e_1$, $v = e_2$, and $w = e_2$. Then $w = 0u + v$, but u is outside Span$\{v, w\}$.

(w) **False.** The hypothesis does not exclude $v_1 = 0$. Take $v_1 = 0$, $v_2 = e_1$, $v_3 = e_2$ in $\mathbb{R}^5$. The second is not a multiple of the first, and the third is outside their span, but the zero vector makes the list dependent.

(x) **True.** A linear transformation is a function with additional operation-preserving properties.

(y) **True.** A 6×5 matrix cannot have six pivots, so its columns cannot span $\mathbb{R}^6$.

(z) **False.** m pivot columns mean onto $\mathbb{R}^m$. For one-to-one we need n pivots. Again [1 0] has $m = 1$ pivot but sends every $(0, t)^T$ to zero.

Exercise 1.S.2: Every possible solution set of one equation

If $a \neq 0$, division gives the unique solution $x = b/a$. If $a = 0$ but $b \neq 0$, the equation says $0 = b$ and has no solution. If $a = b = 0$, the equation is true for every real x, so its solution set is all of $\mathbb{R}$. These three cases exhaust the values of a, b.

Exercise 1.S.3: Three configurations of planes

(a) Use $x = 0$, $y = 0$, $x + y = 0$. The first two force $x = y = 0$, while z is arbitrary; the third adds no restriction. The intersection is the z-axis, a single line.

(b) Use $x = 0$, $y = 0$, $z = 0$. Their only common point is the origin.

(c) Use $x = 0$, $x = 1$, $y = 0$. No point can satisfy the first two simultaneously, so there is no common intersection.

Every listed equation has at least one nonzero coefficient and therefore really does describe a plane, rather than the whole space or an empty set.

Exercise 1.S.4: Why the square size matters

Three pivot columns in a 3×3 coefficient matrix also mean three pivot rows. Thus appending any right-hand side gives a consistent system, because no zero-coefficient row remains to create a contradiction. There are no free variables, so the solution is unique. Both properties are needed: column pivots alone in a taller matrix would not ensure existence.

Exercise 1.S.5: Classify two parameter-dependent systems

(a) Eliminate the first variable.

Subtract four times $x_1 + 3x_2 = k$ from the second equation. The result is $(h - 12)x_2 = 8 - 4k$. If $h \neq 12$, this determines x_2 uniquely, then the first equation determines x_1. If $h = 12$, the result is $0 = 8 - 4k$, consistent exactly when $k = 2$.

Conclusion. (i) No solutions: $h = 12, k \neq 2$. (ii) One solution: $h \neq 12$ with any k. (iii) Infinitely many: $h = 12, k = 2$.

(b) Add three times the first equation to the second.

The eliminated equation is $(k + 3h)x_2 = 1$. If $k + 3h \neq 0$, then $x_2 = 1/(k + 3h)$ and $x_1 = (hx_2 - 1)/2$ are uniquely determined. If $k + 3h = 0$, we obtain the contradiction $0 = 1$.

Conclusion. (i) No solutions: $k = -3h$. (ii) One solution: $k \neq -3h$. (iii) No parameter values give infinitely many solutions.

Exercise 1.S.6: Restate one consistency question three ways

Let $a_1 = (4, 8)^T$, $a_2 = (-2, -3)^T$, $a_3 = (7, 10)^T$, and $b = (-5, -3)^T$.

(a) Use a linear combination.

Ask whether $b = x_1a_1 + x_2a_2 + x_3a_3$ for some real weights. Eliminating twice the first scalar equation from the second gives $x_2 - 4x_3 = 7$. Let $x_3 = t$; then $x_2 = 7 + 4t$ and $4x_1 = 9 + t$. Thus the full solution family is

$$x = (9/4 + t/4,\ 7 + 4t,\ t)^T, \qquad t \in \mathbb{R}.$$

In particular $(9/4, 7, 0)^T$ works, proving consistency.

(b) Use matrix columns.

For $A = \begin{bmatrix} 4 & -2 & 7 \\ 8 & -3 & 10 \end{bmatrix}$, ask whether b lies in the span of the columns of A. Part (a) answers yes.

(c) Use a transformation.

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x) = Ax$. Ask whether b is in the range of T, or equivalently whether any input has image b. The displayed family supplies all such inputs.

Exercise 1.S.7: Restate consistency for every right-hand side

Set $v_1 = (2, -5, 7)^T$, $v_2 = (-4, 1, -5)^T$, and $v_3 = (-2, 1, -3)^T$, and put $A = [v_1\ v_2\ v_3]$.

(a) Ask whether the span is all of $\mathbb{R}^3$.

Row three of A is row one minus row two. Hence every vector b in the column span satisfies $b_3 = b_1 - b_2$. The first two rows have two independent leading columns: elimination of x_1 from them leaves a nonzero coefficient -9 on x_2 after multiplying the first row by $5/2$ and adding it to row two. Thus the first two equations can be solved for any b_1, b_2, but the third requires exactly $b_3 = b_1 - b_2$. Consequently $\mathrm{Span}\{v_1, v_2, v_3\}$ is this plane, not $\mathbb{R}^3$.

(b) State the column version.

The question asks whether the columns of A span $\mathbb{R}^3$. They do not; for example $(0, 0, 1)^T$ is outside their span.

(c) State the transformation version.

For $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x) = Ax$, the question is whether T is onto. It is not, because its range is the proper plane above.

Conclusion. The system is consistent precisely for $b_3 = b_1 - b_2$, so it is not consistent for all triples of right-hand sides.

Exercise 1.S.8: Echelon patterns for spanning columns

Use p, q, r for nonzero pivots and $*$ for arbitrary entries.

(a) Two rows and three columns.

Spanning $\mathbb{R}^2$ requires two row pivots, which can occur in columns $(1, 2)$, $(1, 3)$, or $(2, 3)$:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \end{bmatrix}, \quad \begin{bmatrix} p & * & * \\ 0 & 0 & q \end{bmatrix}, \quad \begin{bmatrix} 0 & p & * \\ 0 & 0 & q \end{bmatrix}.$$

(b) Three rows and three columns.

All three columns must also be pivot columns, leaving the sole pattern $\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}$.

Exercise 1.S.9: Decompose a vector along two different lines

A vector on $y = 2x$ is $(s, 2s)$; one on $y = x/2$ is $(t, t/2)$. Their sum equals $(5, 6)$ when $s + t = 5$ and $2s + t/2 = 6$. Substituting $t = 5 - s$ into the latter and multiplying by two gives $4s + 5 - s = 12$, so $s = 7/3$ and $t = 8/3$. Therefore

$$(5, 6) = (7/3, 14/3) + (8/3, 4/3).$$

Each summand satisfies its respective line equation, and their coordinate sums check the requested result.

Exercise 1.S.10: Infer existence and uniqueness from the diagram

The two source arrows a_1, a_2 are nonzero and not parallel, so they are independent. Their two column directions span $\mathbb{R}^2$. Thus any b shown in that plane is attained, and its two coefficients are unique. Negative coefficients are allowed; being outside the wedge of positive combinations does not prevent a solution of $Ax = b$.

Conclusion. Yes, there is exactly one solution.

Exercise 1.S.11: A nonechelon matrix with a line of zero-images

Choose

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Both rows lead in column one, so it is not in echelon form. The second equation in $Ax = 0$ gives $x_1 = 0$, then the first gives $x_2 = 0$, while x_3 is free. Thus the solution set is $\{(0, 0, t)^T : t \in \mathbb{R}\}$, the required line in $\mathbb{R}^3$.

Exercise 1.S.12: A nonechelon matrix with a plane of zero-images

Choose $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. The second row again leads under the first, so the matrix is not echelon. Its second equation is twice the first, leaving only $x_1 + 2x_2 + 3x_3 = 0$. Its solutions are

$$x = s(-2, 1, 0)^T + t(-3, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

The two generators are independent, so this is a plane, not merely a line.

Exercise 1.S.13: A prescribed null vector determines the missing entries

The pivot information forces reduced echelon form $\begin{bmatrix} 1 & 0 & u \\ 0 & 1 & v \\ 0 & 0 & 0 \end{bmatrix}$. Applying it to $(3, -2, 1)^T$ gives $(3 + u, -2 + v, 0)^T$. For this to be zero, $u = -3$ and $v = 2$. Hence the required form is

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Row operations preserve the null space, so the prescribed vector must satisfy these reduced equations even if the original A was different.

Exercise 1.S.14: Parameter values for two-vector independence

Put the two vectors into columns and replace R_2 by $R_2 - aR_1$:

$$\begin{bmatrix} 1 & a \\ a & a + 2 \end{bmatrix} \sim \begin{bmatrix} 1 & a \\ 0 & a + 2 - a^2 \end{bmatrix}.$$

There are two pivots precisely when $a+2-a^2 \neq 0$. Since $a+2-a^2 = -(a-2)(a+1)$, independence holds for $\boxed{a \neq 2, -1}$. At either excluded value there is one pivot, so a free coefficient supplies dependence.

Exercise 1.S.15: Parameters in triangular-looking vector lists

(a) Three vectors in $\mathbb{R}^3$.

The column matrix is $\begin{bmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{bmatrix}$. Independence requires and is guaranteed by $a, c, f \neq 0$; the other entries b, d, e may be any real numbers. For sufficiency, read a homogeneous relation from bottom to top: $fz = 0$ gives $z = 0$, then $cy = 0$ gives $y = 0$, then $ax = 0$ gives $x = 0$. For necessity, if $a = 0$ the first vector is zero; if $c = 0$, the first two vectors are dependent; if $f = 0$, all three vectors lie in the two-dimensional coordinate plane with third coordinate zero.

(b) Three vectors in $\mathbb{R}^4$.

There are *no restrictions* on a, b, c, d, e, f. A relation among the columns, read from the last coordinate upward, first forces the third coefficient to zero because that coordinate is 1 only in column three. The third coordinate then forces the second coefficient to zero, and the second coordinate forces the first to zero. Hence the list is independent for every choice of the six numbers.

Exercise 1.S.16: Use the preceding-column characterization

The supplied matrix has nonzero diagonal entries $1, 5, 8, 10$ and zeros above the diagonal. Column one is nonzero. For column $j > 1$, suppose it were a combination of the preceding columns. Its first $j-1$ coordinates are zero. Reading those coordinates from top to bottom forces every coefficient of the preceding columns to be zero, because each relevant diagonal entry is nonzero. The proposed combination would then be zero, contradicting the nonzero jth coordinate of column j. Thus no later column is a combination of preceding ones. The characterization in Theorem 7 of Section 1.7 now proves that all four columns are independent.

Exercise 1.S.17: Adjoin a vector outside the old span

Suppose $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$. If $c_4 \neq 0$, solving for v_4 would put it in $\text{Span}\{v_1, v_2, v_3\}$, contrary to the hypothesis. So $c_4 = 0$. Independence of the first three then forces $c_1 = c_2 = c_3 = 0$. All coefficients vanish, proving independence of the four-vector list.

Exercise 1.S.18: Add one basis vector to another

A relation $av_1 + b(v_1 + v_2) = 0$ can be collected as $(a+b)v_1 + bv_2 = 0$. Independence of v_1, v_2 forces $b = 0$ and $a + b = 0$, hence $a = 0$. Thus the new pair $v_1, v_1 + v_2$ is also independent. The argument proves the result for all coefficients, not just a selected numerical example.

Exercise 1.S.19: Three collinear points give dependent position vectors

Because the points are distinct, $v_2 - v_1 \neq 0$ gives a direction for their common line. Some real t satisfies $v_3 = v_1 + t(v_2 - v_1) = (1-t)v_1 + tv_2$. Therefore

$$(1-t)v_1 + tv_2 - v_3 = 0.$$

The coefficient of v_3 is -1, so this relation is nontrivial. It proves dependence even if the line does not pass through the origin: the line's position vectors still lie in the span of two of its points.

Exercise 1.S.20: Reverse an input using homogeneity

Since $-u = (-1)u$, linearity gives $T(-u) = (-1)T(u) = -v$. This uses the scaling rule with the scalar -1 and does not require T to be invertible.

Exercise 1.S.21: A coordinate-plane reflection

The images of e_1, e_2, e_3 are $e_1, -e_2, e_3$. Put them into columns to get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It leaves the first and third coordinates fixed and reverses the second, exactly the specified reflection through $x_2 = 0$.

Exercise 1.S.22: Surjectivity forces injectivity for this square size

Onto $\mathbb{R}^3$ means a pivot in every row of the 3×3 matrix. These three pivots also occupy every column, so there is no free variable in $Ax = 0$. Its only solution is zero, giving one-to-one. The equality of the input and output dimensions is crucial here.

Exercise 1.S.23: A Givens rotation in two coordinates

Multiplying the displayed matrix by $(4, 3)^T$ gives $(4a - 3b, 4b + 3a)^T$. The target requires $4a - 3b = 5$ and $3a + 4b = 0$. From the second equation, $b = -3a/4$; substituting in the first gives $25a/4 = 5$. Thus $\boxed{a = 4/5, b = -3/5}$. The condition $a^2 + b^2 = 16/25 + 9/25 = 1$ is satisfied, and the image is $(5, 0)^T$, so both direction and length check.

Exercise 1.S.24: Zero one coordinate without changing the middle one

The first and third coordinates give $2a - 4b = 2\sqrt{5}$ and $2b + 4a = 0$. The latter says $b = -2a$, so $10a = 2\sqrt{5}$. Therefore

$$a = 1/\sqrt{5}, \qquad b = -2/\sqrt{5}.$$

The middle coordinate remains 3 automatically, and $a^2 + b^2 = 1/5 + 4/5 = 1$. Substituting in the first and third coordinates gives $2\sqrt{5}$ and 0, as required.

Exercise 1.S.25: A building design needs whole, nonnegative floor counts

(a) Interpret a scalar multiple of the recipe column.

The vector $x_1(3,7,8)^T$ gives the numbers of three-, two-, and one-bedroom apartments on the x_1 floors using plan A.

(b) Add the contributions from all three plans.

The total apartment-count vector is

$$x_1\begin{bmatrix}3\\7\\8\end{bmatrix}+x_2\begin{bmatrix}4\\4\\8\end{bmatrix}+x_3\begin{bmatrix}5\\3\\9\end{bmatrix}.$$

(c) Solve the target equations, then impose the physical restrictions.

Appending $(66,74,136)^T$ and reducing gives

$$\left[\begin{array}{ccc|c}3&4&5&66\\7&4&3&74\\8&8&9&136\end{array}\right]\sim\left[\begin{array}{ccc|c}1&0&-1/2&2\\0&1&13/8&15\\0&0&0&0\end{array}\right].$$

Thus the real solutions are $x_1=2+t/2$, $x_2=15-13t/8$, $x_3=t$. Floors must be nonnegative integers. Integer t makes x_2 an integer only when t is a multiple of 8, since 13 and 8 have no common factor. Nonnegativity requires $0\le t\le 120/13<10$. Hence the only allowed values are $t=0,8$, giving $(x_1,x_2,x_3)=(2,15,0)$ or $(6,2,8)$. Substitution yields the requested apartment totals in both cases.

Conclusion. Yes. There are exactly two designs with whole nonnegative floor counts, although the unrestricted real system has infinitely many solutions.